

Hotel Booking Dataset — Executive EDA Report

End-to-end exploratory analysis, data-quality assessment, business insights and management recommendations

Metric	Result
Raw records	25,180
Raw columns	35
Duplicate rows removed	180
Cleaned records	25,000
Cancellation rate	72.2%
Average ADR	140.59
Average stay	3.69 nights
Total estimated revenue	13,384,199.45
Average satisfaction	3.85/5

1. Data Quality & Preprocessing

The dataset contains 25,180 rows and 35 columns. The quality review identified 180 exact duplicate rows, missing values in fields such as Company_ID, Country, Agent_ID, Meal_Type, Satisfaction_Score, Children, Hotel_Location and ADR, plus inconsistent categorical spellings/capitalization. Dates were converted to datetime. The single satisfaction score above the valid 0–5 scale was treated as invalid. The one Total_Nights inconsistency was corrected from weekend plus weekday nights. Missing identifiers were labeled Unknown; numeric missing values were median-imputed. Statistical outliers were flagged using IQR rather than automatically removed because extreme bookings can be commercially meaningful.

2. Descriptive Findings

Average lead time is 182.2 days and the median is 182 days. Average stay is 3.69 nights. Average ADR is 140.59, while estimated revenue averages 535.37 per booking. The cancellation rate is 72.2%, making cancellation management the most important commercial theme in the analysis.

3. Key Group-wise Results

Hotel Type	Bookings	Cancel %	Avg ADR	Avg Revenue
Resort Hotel	9,389.0	70.3%	173.56	784.52
City Hotel	15,611.0	73.3%	120.76	385.52

4. Key Business Insights

1. Cancellations are exceptionally high at 72.2%, creating substantial revenue uncertainty.
2. Non Refund bookings show about 93.0% cancellations versus 51.6% for No Deposit bookings.
3. Resort Hotel averages about 173.56 ADR versus 120.76 for City Hotel and generates higher average revenue per booking.
4. Online TA is the largest normalized market segment with 10,852 bookings and a 75.1% cancellation rate.
5. ADR and stay length are key revenue drivers; management should coordinate pricing and length-of-stay controls.

5. Management Recommendations

1. Deploy cancellation-risk scoring and stronger confirmation/payment milestones for high-risk, long-lead and high-value reservations.

2. A/B test partial-deposit and flexible-protection policies to reduce cancellations without unnecessarily suppressing conversion.
3. Reduce OTA dependency through direct-booking benefits, channel-level contract reviews and targeted re-confirmation campaigns.
4. Apply hotel-type-specific revenue management, with stronger yield optimization on resort inventory.
5. Create destination-level pricing and promotion rules based on booking, ADR and cancellation patterns.
6. Build an exception dashboard for extreme ADR, revenue, lead time and waiting-list observations instead of deleting them.
7. Link satisfaction, special requests and repeat-guest behavior to service improvements and direct-channel retention.

6. Deliverables

The accompanying Jupyter/Google Colab notebook contains reproducible Python, Pandas, Matplotlib and Seaborn analysis, including data-quality checks, cleaning, visualizations, group-wise analysis, correlation analysis, insights and recommendations. A cleaned CSV is also provided for downstream work.